Theory and Methods for Certifying Deep Generative Models

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Abstract

Deep generative models can produce high-quality samples, but empirical evaluation does not guarantee that learned distributions are close to unknown target distributions. Recent work derived a finite-sample, high-probability upper bound on the Wasserstein distance between the target and the distribution learned by a denoising diffusion probabilistic model (DDPM). This project investigates the numerical reproducibility and extensibility of this certificate.

The original two-dimensional uniform-square experiment was replicated using the authors' released implementation. Across the same six values of the trade-off parameter λ, the published bounds were reproduced with relative differences of at most 3.34%. The framework was then extended to three bounded two-dimensional distributions with different geometries: two moons, checkerboard, and circles. For each distribution, a DDPM was trained and the total bound and its five constituent terms were evaluated across the same λ settings, alongside comparisons of generated and target samples.

Across the extensions, bounds for $\lambda \in \{n/10, n/5, n/2, n\}$ ranged from 1.14 to 2.28, below the $\sqrt{8} \approx 2.83$ trivial bound for the shared $[-1, 1]^2$ instance space, whereas the two larger λ settings produced vacuous bounds from 3.04 to 11.28. Reconstruction error was the largest component throughout the informative settings, while the increasing diameter penalty drove the loss of non-vacuity at larger λ. These results show that the certificate is reproducible and can remain informative across several low-dimensional geometries, while revealing how its usefulness depends on parameterisation and bound composition.

Declaration

No portion of the work referred to in this dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

Copyright Statement

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1 Introduction

1.1 Background and Motivation

Deep generative models aim to learn the underlying probability distribution of observed data and to generate new samples resembling those from the original data-generating process. Their ability to represent high-dimensional, complex distributions has driven substantial progress in areas such as image synthesis, data augmentation and scientific data generation. Among the principal families of deep generative models, denoising diffusion probabilistic models have attracted particular attention for their ability to generate high-quality samples through a sequence of learned denoising operations (Ho, Jain and Abbeel, 2020).

Despite this empirical success, reliably evaluating deep generative models remains challenging. Common approaches include visual inspection of generated samples and empirical measures such as the Fréchet Inception Distance (FID), which compares feature-space statistics of real and generated samples (Heusel et al., 2018). Like visual inspection, however, FID is a summary statistic computed from a finite batch of samples rather than a formal guarantee. In particular, finite-sample estimates of FID are biased, with the bias depending on both the sample size and the model being evaluated (Chong and Forsyth, 2020). It does not itself certify, with any stated confidence, how far the learned distribution lies from the unknown target distribution. Moreover, it depends on the chosen feature representation, the number of evaluation samples and the dataset considered. Strong empirical performance therefore does not necessarily establish that a generative model has accurately learned the target distribution.

This limitation motivates the study of certification for deep generative models. In this context, a certificate is a high-probability, data-dependent bound that quantifies the discrepancy between the target and learned distributions at a stated confidence level. Certification has been studied more extensively in supervised learning, where PAC-Bayesian methods have produced non-vacuous risk certificates for probabilistic neural networks (Pérez-Ortiz et al., 2021). Extending comparable guarantees to generative modelling is harder, however, since the object being learned is an entire distribution rather than a predictor evaluated through classification error. Recent work by Mbacke and Rivasplata (2024) makes progress towards this objective by deriving a quantitative upper bound on the Wasserstein distance between a data-generating distribution and the distribution learned by a diffusion model, providing a principled complement to sample-based and heuristic evaluation methods. Its practical value, however, depends on whether the bound can be computed reliably, whether it produces informative numerical values, and whether it extends beyond simple experimental settings. These questions motivate the present project: to reproduce the original certification experiment, evaluate the numerical behaviour of the resulting bound, and investigate its extension to further data distributions.

1.2 Research Gap and Problem Statement

PAC-Bayesian research has demonstrated that informative, non-vacuous certificates can be obtained for supervised neural networks, but these results concern prediction risk rather than the quality of a learned generative distribution (Pérez-Ortiz et al., 2021). For diffusion models, Mbacke and Rivasplata (2024) derived a finite-sample, high-probability upper bound on the Wasserstein distance between the target and learned distributions. However, the accompanying empirical evidence was limited to a two-dimensional uniform distribution, leaving the practical behaviour of the bound insufficiently explored.

Three limitations remain particularly relevant. First, it is unknown whether the reported numerical behaviour, including the non-vacuity and sensitivity of the bound to λ, extends to target distributions with different geometry or greater complexity. Second, the bound does not generally converge to zero as the sample size tends to infinity, indicating a potential limitation in the asymptotic tightness of the certificate, rather than necessarily of the diffusion model itself. Third, the numerical analysis estimates the required Lipschitz constants using the analytical quantity K'_t, but the relationship between this estimate and the effective Lipschitz behaviour of the trained network is not empirically examined. Consequently, the applicability and tightness of the certificate beyond the original synthetic setting remain open, including the question of which new datasets or target distributions provide appropriate and informative settings for further evaluation.

1.3 Certification and PAC-Bayesian Learning

The problem of certifying a learned model — producing a numerical guarantee, valid with a stated confidence, on a quantity that cannot be observed directly — has a long history in statistical learning theory. Early approaches bounded the gap between empirical and population risk using measures of hypothesis-class complexity, including VC dimension and Rademacher or Gaussian complexities (Vapnik, 1998; Bartlett and Mendelson, 2002). For modern neural networks these proved uninformative: Zhang et al. (2016) showed that standard architectures can fit arbitrary labellings of their training data, so any bound derived from class capacity alone must be extremely loose, and Neyshabur et al. (2017) reached similar conclusions across several candidate complexity measures. The resulting bounds were typically vacuous, exceeding the trivial worst-case value of the loss and therefore saying nothing about the trained model. Certification consequently shifted towards bounds that depend on the training data and on the specific predictor obtained, rather than on the size of the hypothesis class alone.

PAC-Bayesian analysis emerged as one of the most successful such approaches, certifying a distribution over hypotheses by bounding its population risk through empirical risk and a Kullback–Leibler divergence from a prior fixed independently of the certification data (Shawe-Taylor and Williamson, 1997; McAllester, 1999), with subsequent refinements tightening the dependence on sample size (Maurer, 2004). A major empirical breakthrough came from Dziugaite and Roy (2017), who obtained the first non-vacuous bounds for stochastic networks with far more parameters than training examples by optimising the bound itself rather than evaluating it post hoc. Zhou et al. (2019) extended non-vacuous guarantees to ImageNet-scale architectures via compression. Pérez-Ortiz et al. (2021) then shifted attention from validity to tightness, training probabilistic networks on objectives derived from tight PAC-Bayes bounds and reporting certificates close to the observed test error — a distinction that matters because a bound may be correct yet too loose to be useful.

These results establish that informative numerical certification is achievable in supervised deep learning, and that the usefulness of a certificate depends not only on its formal validity but on whether its numerical value is small enough to be informative — a property that had to be pursued deliberately rather than obtained for free. Transferring this to generative modelling is not immediate, because the certified quantity is a discrepancy between two distributions rather than a classification risk, and the model is a multi-step stochastic sampler rather than a predictor. Mbacke, Clerc and Germain (2023) took a first step by deriving PAC-Bayesian guarantees for variational autoencoders, expressing the generation term as a Wasserstein distance; Mbacke and Rivasplata (2024) subsequently adapted this construction to DDPMs. For diffusion models, however, the corresponding numerical question remains largely unexamined: a valid quantitative bound now exists, but its behaviour across settings, and the conditions under which it stays informative, have received far less scrutiny than their supervised counterparts.

1.4 Distributional Guarantees for Diffusion Models

A central question in certifying a generative model is therefore which discrepancy should be used to compare the target and learned distributions. Common choices include Kullback–Leibler divergence, total variation distance and Wasserstein distance. These quantities capture different notions of distributional similarity. KL divergence may become infinite when the required absolute-continuity condition is violated, while total variation distance does not account for the geometry of the sample space. Wasserstein distance is instead defined through an optimal transport cost and, under appropriate moment conditions, can remain finite and informative even when the distributions have disjoint supports (Villani, 2009). This makes it particularly relevant to the evaluation of generative models, although no single discrepancy is universally preferable.

Much of the theoretical literature on diffusion models has analysed convergence through score-based or stochastic-differential-equation formulations. Song et al. (2021) introduced a continuous-time SDE framework that unified score-based generative models and diffusion probabilistic models. Subsequent work established convergence guarantees under assumptions involving score-estimation accuracy, sampling discretisation and properties of the target distribution. For example, Lee, Lu and Tan (2023) obtained Wasserstein convergence guarantees for distributions with bounded support or sufficiently decaying tails under an assumption of L^2-accurate score estimation. Although these results provide important theoretical understanding, assumptions on score-estimation error can make the resulting guarantees difficult to evaluate numerically for a particular trained model.

Mbacke and Rivasplata (2024) developed a different route by deriving a finite-sample, high-probability upper bound on the Wasserstein distance between the target distribution and the distribution learned by a denoising diffusion probabilistic model. Their result applies to arbitrary distributions on bounded instance spaces, does not require the target distribution to admit a density with respect to Lebesgue measure, and avoids assumptions on the learned score function. However, the reported numerical illustration focused on a simple two-dimensional uniform distribution. Consequently, the numerical behaviour, tightness and practical informativeness of the bound in broader experimental settings remain insufficiently understood. These considerations underlie the limitations identified in Section 1.2.

1.5 Aim, Objectives and Research Questions

The project has the following objectives: To reproduce the original two-dimensional diffusion-model experiment and compare the resulting Wasserstein upper bounds with the values reported by Mbacke and Rivasplata (2024).

To investigate how the trade-off parameter λ affects the numerical value and non-vacuity of the certificate.

To identify appropriate new datasets or target distributions for extending the original experiment, considering properties such as bounded support, and data geometry.

To construct a controlled extension in which the model architecture, training configuration and certificate computation are held fixed, so that differences in the resulting bound can be attributed to the target distribution alone.

To provide reproducible code, experimental results and a research report documenting the methodology, findings and limitations of the investigation.

These objectives are organised around three research questions:

RQ1: To what extent can the Wasserstein upper-bound values reported for the original two-dimensional experiment be reproduced?

RQ2: How does the trade-off parameter λ affect the numerical behaviour and non-vacuity of the Wasserstein upper bound?

RQ3: To what extent can the certification method be extended to a new dataset or target distribution while retaining an informative numerical upper bound?

1.6 Contributions

This project makes the following empirical and methodological contributions to the evaluation of Wasserstein-based certificates for diffusion models: It provides a systematic replication of the numerical experiment reported by Mbacke and Rivasplata (2024), including all six values of the trade-off parameter λ, and quantitatively compares the reproduced bounds with the published results.

It examines the effect of λ on the numerical value and non-vacuity of the certificate.

It develops a theoretically informed procedure for selecting an additional dataset or target distribution, accounting for bounded support, dimensionality and data geometry, and adapts the model and certificate evaluation accordingly.

It provides a reproducible experimental implementation that separates model training from certificate evaluation, enabling multiple bound settings to be assessed using the same trained diffusion model.

2 Methodology

2.1 Theoretical Framework

2.1.1 Denoising Diffusion Probabilistic Models

The generative model used in this project is a discrete-time DDPM, based on the diffusion framework of Sohl-Dickstein et al. (2015) and the parameterisation of Ho, Jain and Abbeel (2020). A DDPM can also be interpreted as a Markovian hierarchical variational autoencoder in which the encoder is represented by the fixed forward diffusion process and the decoder corresponds to the learned multi-step reverse process (Luo, 2022). This interpretation provides the methodological connection between the two principal works underlying the present project. Mbacke, Clerc and Germain (2023) first developed PAC-Bayesian reconstruction and Wasserstein generation guarantees for variational autoencoders. Mbacke and Rivasplata (2024) subsequently adapted this framework to DDPMs by treating the terminal forward distribution $q(x_T \mid x_0)$ as the conditional latent distribution and the Gaussian distribution $p(x_T)$ as the latent prior. Let $x_0 \sim \mu$ denote a sample from the target distribution, where $x_0 \in R^D$. The forward process is a Markov chain with transitions

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t)I\right),$$

where $t \in \{1, \ldots, T\}$, $\alpha_t = 1 - \beta_t$, and β_t controls the amount of noise introduced at time step t. Defining

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s,$$

the distribution of x_t conditioned directly on x_0 has the closed form

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) I\right).$$

Consequently, a noisy sample can be generated without simulating all preceding time steps:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

For a sufficiently long and appropriately selected noise schedule, the terminal distribution $q(x_T \mid x_0)$ becomes close to the standard Gaussian prior $p(x_T) = \mathcal{N}(0, I)$. The learned reverse process attempts to invert this transformation. Its transitions are represented as

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; g_\theta^t(x_t), \sigma_t^2 I\right),$$

where g_θ^t is a neural-network-dependent mean function and σ_t^2 is determined by the variance schedule. Following Ho, Jain and Abbeel (2020), the neural network can be parameterised to predict the Gaussian noise ϵ that produced x_t. During training, a data sample, a time step and a noise vector are sampled, after which the model minimises the simplified noise-prediction objective

$$\mathcal{L}_{\mathrm{DDPM}}(\theta) = E_{x_0, t, \epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right].$$

After training, generation begins with $x_T \sim \mathcal{N}(0, I)$ and repeatedly applies the learned reverse transitions until an approximate data sample x_0 is obtained. All subsequent experiments and certificate calculations therefore use this discrete-time DDPM formulation. In the replicated two-dimensional experiment, the noise-prediction function is implemented using a fully connected neural network conditioned on sinusoidal embeddings of the noisy coordinates and the diffusion time step; the detailed architecture and training configuration are presented in Section 2.2.

2.1.2 Wasserstein Distance

In this project, the discrepancy between the target distribution μ and the distribution learned by the model π_θ is quantified using the first-order Wasserstein distance $W_1(\mu, \pi_\theta)$. This metric is adopted because it incorporates the geometry of the underlying data space, so the discrepancy depends not only on differences in probability mass but also on the distance over which that mass must be transported (Villani, 2009; Arjovsky, Chintala and Bottou, 2017).

Let p and q be probability distributions on a metric space $\mathcal{X}$. A coupling γ of p and q is a joint distribution on $\mathcal{X} \times \mathcal{X}$ whose marginal distributions are p and q, respectively. The set of all such couplings is denoted by $\Gamma(p, q)$. For distributions on R^D equipped with the Euclidean metric, the Wasserstein distance of order one is defined as

$$W_1(p, q) = \inf_{\gamma \in \Gamma(p,q)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|_2 \, d\gamma(x, y).$$

Intuitively, $W_1(p, q)$ represents the minimum expected transportation cost required to transform probability mass distributed according to p into probability mass distributed according to q, where the cost of transporting mass from x to y is their Euclidean distance. Unlike total variation distance, the Wasserstein distance accounts for how far apart samples are in the underlying space. It can also remain finite and informative when two distributions have disjoint supports, whereas the Kullback–Leibler divergence may be infinite when the required absolute-continuity condition is not satisfied (Villani, 2009; Mbacke and Rivasplata, 2024).

The Wasserstein distance is a metric and therefore satisfies the triangle inequality. This property is central to the certification method used in this project. Denoting the target distribution by μ, the distribution learned by the diffusion model by π_θ, and a sample-dependent regenerated distribution by μ_θ^n, the target discrepancy can be decomposed as

$$W_1(\mu, \pi_\theta) \leq W_1(\mu, \mu_\theta^n) + W_1(\mu_\theta^n, \pi_\theta).$$

The two terms can then be controlled separately using the empirical reconstruction loss, prior matching and the Lipschitz properties of the reverse diffusion process. The boundedness of the instance space also provides a baseline for interpreting the resulting certificate. If both distributions are supported on a space with finite diameter

$$\Delta = \sup_{x,x' \in \mathcal{X}} \|x - x'\|_2,$$

then the product coupling gives $W_1(p, q) \leq \Delta$. In the replicated experiment, $\mathcal{X} = [-1, 1]^2$, and hence

$$\Delta = \sqrt{(2)^2 + (2)^2} = \sqrt{8} \approx 2.828.$$

A computed upper bound below $\sqrt{8}$ therefore improves upon this direct bounded-space baseline and is treated as non-vacuous in the original numerical experiment.

2.1.3 Lipschitz Continuity

The Wasserstein certificate used in this project requires each reverse-process mean function to satisfy a Lipschitz continuity condition (Mbacke and Rivasplata, 2024). Let $g : \mathcal{X} \to \mathcal{Y}$ be a function between Euclidean spaces. The function is K-Lipschitz continuous if there exists a finite constant $K \geq 0$ such that

$$\|g(x) - g(y)\|_2 \leq K\|x - y\|_2 \qquad \text{for all } x, y \in \mathcal{X}.$$

The smallest constant satisfying this condition is called the Lipschitz constant of g. It bounds the maximum extent to which a difference between two inputs can be amplified by the function; in particular, $K < 1$ implies that the function is contractive (Gouk et al., 2021). For the reverse diffusion process, Mbacke and Rivasplata (2024) assume that the mean function g_θ^t at every time step is K_θ^t-Lipschitz continuous:

$$\left\|g_\theta^t(x) - g_\theta^t(y)\right\|_2 \le K_\theta^t \|x - y\|_2.$$

This assumption is required because a diffusion model applies a sequence of reverse transitions rather than a single mapping. The Lipschitz constant of a composition is bounded by the product of the constants of its component functions (Gouk et al., 2021). Consequently, a difference introduced at the terminal diffusion state may propagate through the complete reverse process according to products of the form

$$\prod_{t=1}^{T} K_\theta^t.$$

Following the Gaussian parameterisation of the DDPM reverse process, a transition at $t \ge 2$ can be represented as

$$x_{t-1} = g_\theta^t(x_t) + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

(Ho, Jain and Abbeel, 2020). For two independently sampled reverse transitions, the Lipschitz condition and the triangle inequality give

$$E\|x_{t-1} - y_{t-1}\|_2 \le K_\theta^t \|x_t - y_t\|_2 + \sigma_t E\|\epsilon - \epsilon'\|_2$$

(Mbacke and Rivasplata, 2024, Lemma 3.3). Repeated application of this inequality produces two components of the final certificate. The product of the Lipschitz constants controls how the discrepancy between the terminal forward distribution $q(x_T \mid x_0)$ and the Gaussian prior $p(x_T)$ propagates through the reverse process. A second, weighted sum controls the accumulation of Gaussian sampling noise across the reverse steps (Mbacke and Rivasplata, 2024, Lemmas 3.4–3.5). The true constants K_θ^t of the trained neural network are not calculated directly in the original numerical experiment. Instead, Mbacke and Rivasplata (2024, Remark 3.2) use the Lipschitz constant of the corresponding forward posterior mean as a proxy:

$$K_t' = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}.$$

For $0 < \alpha_t < 1$, this quantity satisfies $K_t' < 1$. Its use is motivated by the fact that the learned reverse mean is trained to approximate the forward posterior mean. Nevertheless, K_t' is a schedule-derived proxy rather than a measured Lipschitz constant of the trained network. The same approximation is retained when computing the Lipschitz-dependent terms in the replicated certificate.

2.1.4 Regenerated Distribution and Reconstruction Loss

This project uses the regenerated distribution to connect reconstruction from observed data with generation from the diffusion prior. For a sample x_0, define

$$\pi_\theta(\cdot \mid x_0) = q(x_T \mid x_0) p_\theta(x_{T-1} \mid x_T) \cdots p_\theta(\cdot \mid x_1).$$

This is the distribution obtained by first sampling x_T from the terminal forward distribution conditioned on x_0, and then passing it through the complete learned reverse process. Given an i.i.d. sample $S = \{x_0^1, \ldots, x_0^n\}$, the regenerated distribution is

$$\mu_\theta^n = \frac{1}{n} \sum_{i=1}^{n} \pi_\theta(\cdot \mid x_0^i).$$

Unlike the learned distribution π_θ, which begins from the Gaussian prior $p(x_T)$, μ_θ^n begins from noisy versions of observed samples. The associated reconstruction loss is

$$\ell_\theta(x_T, x_0) = E_{p_\theta(\hat{x}_0 \mid x_T)} \left[\|x_0 - \hat{x}_0\|_2 \right],$$

where the expectation represents the complete reverse process.

2.1.5 PAC-Bayesian-Style Analysis

This project follows the PAC-Bayesian-style analysis used by Mbacke and Rivasplata (2024) to control the distance between the target distribution μ and the regenerated distribution μ_θ^n. In this construction, the data-dependent terminal distribution $q(x_T \mid x_0)$ plays a role analogous to a posterior distribution, while the Gaussian distribution $p(x_T)$ acts as a data-independent prior. Their discrepancy is therefore measured by

$$\mathrm{KL}(q(x_T \mid x_0) \,\|\, p(x_T)) \,.$$

Combining this prior-matching term with the empirical reconstruction loss gives, with probability at least $1-\delta$,

$$W_1(\mu, \mu_\theta^n) \leq \frac{1}{n}\sum_{i=1}^{n} E_{q(x_T|x_0^i)}\ell_\theta(x_T, x_0^i) + \frac{1}{\lambda}\left[\sum_{i=1}^{n} \mathrm{KL}\big(q(x_T \mid x_0^i) \| p(x_T)\big) + \log\frac{1}{\delta}\right] + \frac{\lambda\Delta^2}{8n}.$$

Unlike conventional PAC-Bayes analysis, this construction concerns distributions over terminal latent variables rather than distributions over network weights.

2.1.6 PAC-Bayesian-Style Wasserstein Bound

This project uses the finite-sample Wasserstein certificate derived by Mbacke and Rivasplata (2024). If $\mathcal{X}$ has finite diameter Δ, then for $\lambda > 0$ and $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\begin{aligned} W_1(\mu, \pi_\theta) \leq{}& \frac{1}{n}\sum_{i=1}^{n} E_{q(x_T|x_0^i)}\ell_\theta(x_T, x_0^i) + \frac{1}{\lambda}\left[\sum_{i=1}^{n} \mathrm{KL}\big(q(x_T \mid x_0^i) \,\|\, p(x_T)\big) + \log\frac{1}{\delta}\right] + \frac{\lambda\Delta^2}{8n} \\ &+ \left(\prod_{t=1}^{T} K_\theta^t\right)\frac{1}{n}\sum_{i=1}^{n} E_{q(x_T|x_0^i)} E_{p(y_T)}\|x_T - y_T\|_2 + \left[\sum_{t=2}^{T}\left(\prod_{j=1}^{t-1} K_\theta^j\right)\sigma_t\right] E_{\epsilon,\epsilon'}\|\epsilon - \epsilon'\|_2. \end{aligned} \tag{1}$$

The terms respectively represent reconstruction error, prior matching and confidence, the bounded-space penalty, propagated terminal mismatch, and accumulated reverse-process noise. This project computes their sum as the estimated numerical certificate for $W_1(\mu, \pi_\theta)$.

2.2 Replication of the Original Experiment

2.2.1 Code Provenance and Scope of the Replication

The replication used the authors' supplementary implementation for Mbacke and Rivasplata (2024), obtained from the paper's OpenReview submission. The original data generation, diffusion model, training procedure and Wasserstein certificate calculations were retained unchanged. Modifications were limited to experiment management, including model saving/loading, automated result recording and evaluation of multiple λ values using the same trained model. The aim was therefore to reproduce the published numerical results using the supplied implementation rather than to reimplement or modify the method.

2.2.2 Experimental Data

The target distribution was the uniform distribution on a two-dimensional square of side length two centred at the origin:

$$\mu = \mathcal{U}([-1, 1]^2).$$

The data were generated synthetically by sampling the two coordinates independently from $\mathcal{U}(-1, 1)$. The resulting instance space was $\mathcal{X} = [-1, 1]^2$, with Euclidean diameter

$$\Delta = \sqrt{(2)^2 + (2)^2} = \sqrt{8}.$$

A training set containing 50,000 samples was generated for the model optimisation. A separate set of $n = 5,000$ samples was generated after training and used exclusively for certificate evaluation. A further 2,000 target samples and 2,000 model-generated samples were used for qualitative visualisation but did not contribute to the certificate calculation.

2.2.3 Model Architecture

The generative model was a discrete-time DDPM with $T = 1,000$ diffusion steps. A linear variance schedule was used, increasing from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. A single shared noise-prediction network was applied at every diffusion step. Its inputs were the two coordinates of the noisy sample, $x_{t,1}$ and $x_{t,2}$, and the diffusion time step t. Each scalar input was mapped independently to a 128-dimensional sinusoidal embedding. The three embeddings were concatenated to produce a 384-dimensional input vector. The network consisted of four fully connected hidden layers. The first layer mapped the 384-dimensional input to 128 hidden units, followed by three $128 \to 128$ layers. Each hidden layer used a GELU activation. The final $128 \to 2$ linear layer predicted the two-dimensional Gaussian noise vector. The network contained 99,074 trainable parameters.

2.2.4 Training Procedure

The model was trained using 50,000 synthetically generated samples. The training `DataLoader` was configured with a batch size of 100, and optimisation was performed for 500 epochs. For each training example, a diffusion time step t and a Gaussian noise vector $\epsilon \sim \mathcal{N}(0, I)$ were sampled independently, and the noisy observation x_t was generated directly from the corresponding clean sample using the precomputed diffusion schedule.

The network was trained to predict the sampled Gaussian noise from (x_t, t). The training objective was the mean-squared error between the true and predicted noise. Optimisation used AdamW with a learning rate of 10^{-4}, with the remaining optimiser settings left at their PyTorch defaults.

After training, the parameters of the reverse-process network were saved to disk. This trained model was subsequently reused for certificate evaluation, so no retraining was performed when evaluating different values of λ.

No explicit random seed was set in either the training or the evaluation script. In addition, the synthetic datasets were generated using NumPy's `default_rng()` without a specified seed, so a fresh certification set was drawn each time the evaluation script was run. The replication was therefore stochastic rather than fully deterministic.

2.2.5 Certificate Evaluation Procedure

The bound was evaluated separately from training, with the loaded model parameters held fixed, on a new independent certification set of $n = 5{,}000$ samples, using $\delta = 0.05$ and $\Delta = \sqrt{8}$. The reconstruction component was estimated using one terminal draw from $q(x_T \mid x_0)$ and one stochastic reverse trajectory per certification sample, while the prior-matching term was evaluated analytically. The Lipschitz quantities were computed using the schedule-derived expression implemented in the authors' supplied code, with the first value set to unity as specified in the implementation. The expected Gaussian-distance terms were estimated by Monte Carlo using 10^6 draws.

The reconstruction, KL sum, propagated terminal-mismatch and accumulated-noise quantities were computed once and reused across $\lambda \in \{n/10,\ n/5,\ n/2,\ n,\ n/0.5,\ n/0.1\}$. Across these evaluations, only the two PAC-Bayesian terms involving λ, namely $(\mathrm{KL} + \log(1/\delta))/\lambda$ and $\lambda\Delta^2/(8n)$, changed. Theorem 3.1 provides a high-probability guarantee for each fixed λ rather than a simultaneous guarantee over all six values; the resulting values were therefore treated as separate certificates.

2.2.6 Replication Setting

A comparison between the theoretical description and the released implementation identified several implementation-level differences. First, although the training DataLoader is configured with a batch size of 100, the supplied training loop applies `batch[0]` before computing the loss, so the effective update does not use the full minibatch as configured. Second, in the accumulated reverse-process noise term, σ_t is multiplied within the product over preceding Lipschitz factors rather than outside it, and is implemented using the forward variance β_t. Third, the implemented Lipschitz expression uses $\sqrt{\bar{\alpha}_t}$, whereas Remark 3.2 specifies $\sqrt{\alpha_t}$. These behaviours were retained unchanged because the aim was to reproduce the released implementation rather than construct a corrected implementation of the theoretical bound.

2.3 Extension

2.3.1 Dataset Selection and Rationale

The extension datasets were selected to vary the structure of the target distribution while preserving the main conditions of the original experiment. All experiments retain the two-dimensional ambient space and common bounded instance space $\mathcal{X} = [-1, 1]^2$, giving the same conservative diameter $\Delta = \sqrt{8}$. This preserves the finite-diameter assumption of Theorem 3.1 (Mbacke and Rivasplata, 2024) while avoiding simultaneous changes in dimensionality and distribution, so differences in the resulting bounds can be interpreted more directly in relation to the target distribution.

The datasets form a simple 2×2 comparison along two structural dimensions: whether the target is full-dimensional with a two-dimensional Lebesgue density or supported on a lower-dimensional set without such a density, and whether its support is relatively simple or more structurally complex. This distinction between ambient and intrinsic dimensionality is consistent with the manifold hypothesis, under which data may concentrate near lower-dimensional structures (Fefferman, Mitter and Narayanan, 2013). The original uniform square provides the full-dimensional, connected baseline, while the checkerboard remains full-dimensional but introduces a fragmented, multimodal spatial structure.

The circle provides a contrasting lower-dimensional case. Samples are drawn uniformly from the circumference without additive noise, giving a one-dimensional support embedded in R^2 with no density relative to two-dimensional Lebesgue measure. This directly examines the paper's claim that the certificate also applies to such distributions (Mbacke and Rivasplata, 2024). Maintaining a noise-free construction is important because adding independent continuous noise with a density can make a lower-dimensional distribution absolutely continuous (Arjovsky and Bottou, 2017).

Two moons combines lower-dimensional support with more complex geometry. Two noise-free curved components are translated and rescaled to remain within $[-1, 1]^2$, preserving the common instance space and the absence of a two-dimensional Lebesgue density. It therefore introduces curved, non-convex and disconnected structure without changing the ambient dimensionality.

Together, square and checkerboard provide full-dimensional targets of differing spatial complexity, while circle and two moons provide noise-free lower-dimensional targets that differ in geometry and connectedness. This controlled design allows the numerical behaviour of the certificate to be compared across structurally distinct distributions while keeping the principal experimental setting fixed.

Table 1: The 2×2 design of target distributions

	Simple / connected	Fragmented / disconnected
Full-dimensional, Lebesgue density	Square	Checkerboard
Lower-dimensional, no density	Circle	Two moons

2.3.2 Extension Implementation and Certificate Computation

The extension experiments retained the same diffusion model, training configuration and certificate calculation procedure used in the replication. The main changes were therefore limited to the target-data generators and one adjustment to the training data pipeline. Separate diffusion models were trained for the circle, checkerboard and two-moons distributions using the same architecture and hyperparameters described in Section 2.2, with 50,000 training samples and an independent certification set of 5,000 samples for each dataset. The common instance space remained $\mathcal{X} = [-1, 1]^2$, with $\Delta = \sqrt{8}$ and $\delta = 0.05$ unchanged.

The original uniform-square generator was replaced by dataset-specific sampling functions. Circle samples were generated from $(\cos\theta, \sin\theta)$ with $\theta \sim U(0, 2\pi)$. The checkerboard distribution was generated by sampling uniformly from alternating cells of a 4×4 grid over $[-1, 1]^2$. The two-moons distribution was generated from two semicircular components and then translated and rescaled so that the complete noise-free support remained within the same bounded instance space.

A small change was also made to the training `DataLoader`. In the released implementation, the underlying tensor was passed directly to the loader, while the training routine subsequently applies `batch = batch[0]`. This causes the first observation of each nominal minibatch to be selected rather than the full minibatch. The extension scripts instead pass the `TensorDataset` directly to the `DataLoader`, so that `batch[0]` correctly unwraps the complete batch of 100 observations. The remaining training logic was unchanged.

For certificate evaluation, the same six values of λ used in the replication were retained. In addition to computing the final Wasserstein upper bound, a separate script was introduced to export the individual bound components. The reconstruction term, prior-matching quantity, terminal mismatch and accumulated reverse-process noise were computed once, while the λ-dependent prior/confidence and diameter terms were evaluated separately for each λ. Their sum reproduced the original certificate calculation while allowing the contribution of each term to be analysed across datasets.

3 Result and Evaluation

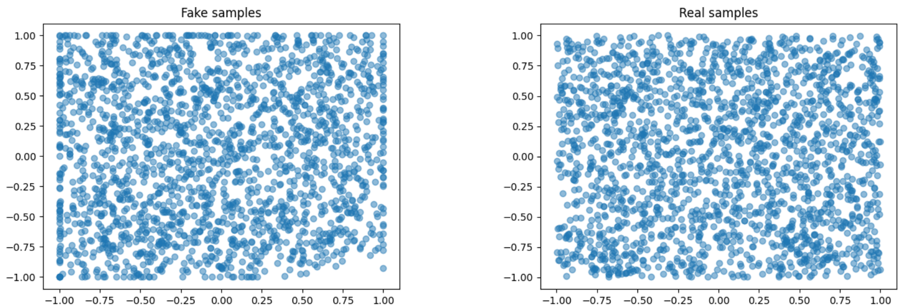
3.1 Replication of the Original Experiment

3.1.1 Replication 1

Table 2: Bound terms and comparison with the paper bound for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Paper bound	Absolute difference	Relative difference (%)
$n/10$	1.055253267	0.006124653	0.1	0	0.000212675	1.161590576	1.124	0.037590576	3.344357311%
$n/5$	1.055253267	0.003062326	0.2	0	0.000212675	1.258528352	1.231	0.027528352	2.236259284%
$n/2$	1.055253267	0.001224931	0.5	0	0.000212675	1.556690812	1.5181	0.038590812	2.542046776%
n	1.055253267	0.000612465	1	0	0.000212675	2.056078434	2.035	0.021078434	1.035795282%
$n/0.5$	1.055253267	0.000306233	2	0	0.000212675	3.055772305	3.056	0.000227695	0.007450768%
$n/0.1$	1.055253267	0.000061247	10	0	0.000212675	11.055527687	11.061	0.005472313	0.049473944%

3.1.1.1 Reproduction of the Published Bound Values



(a) Generated samples (b) Real samples

Figure 1: Comparison between generated and real samples

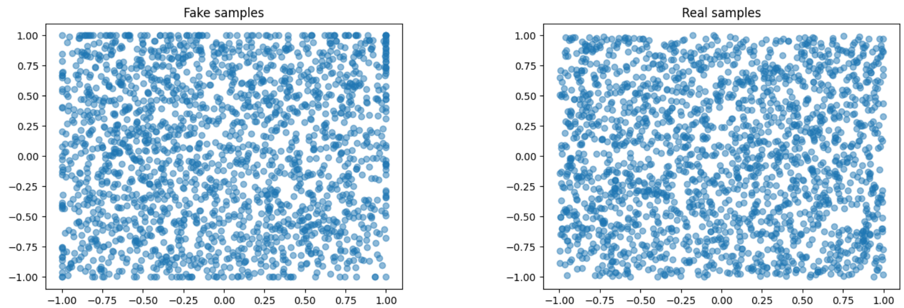
3.1.1.2 Comparison of Generated Samples

3.1.2 Replication 2

Table 3: Bound terms and comparison with the paper bound for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Paper bound	Absolute difference	Relative difference (%)
$n/10$	1.054902554	0.006126788	0.1	0	0.000212518	1.161241889	1.124	0.037241889	3.313335320%
$n/5$	1.054902554	0.003063394	0.2	0	0.000212518	1.258178592	1.231	0.027178592	2.207846607%
$n/2$	1.054902554	0.001225358	0.5	0	0.000212518	1.556340456	1.5181	0.038240456	2.518968185%
n	1.054902554	0.000612679	1	0	0.000212518	2.055727720	2.035	0.020727720	1.018561192%
$n/0.5$	1.054902554	0.000306339	2	0	0.000212518	3.055421352	3.056	0.000578648	0.018934804%
$n/0.1$	1.054902554	0.0000613	10	0	0.000212518	11.05517673	11.061	0.005823265	0.052646823%

3.1.2.1 Reproduction of the Published Bound Values



(a) Generated samples

(b) Real samples

Figure 2: Comparison between generated and real samples

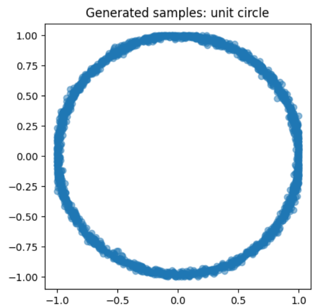
3.1.2.2 Comparison of Generated Samples

3.2 Extension

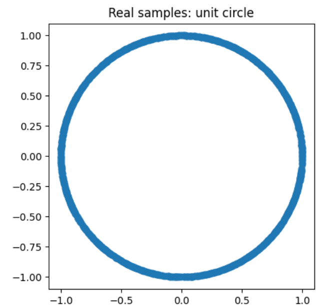
3.2.1 Circle

Table 4: Bound terms for the unit-circle dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.277906	0.006193	0.1	0	0.000213	1.384312	Yes
$n/5$	1.277906	0.003097	0.2	0	0.000213	1.481215	Yes
$n/2$	1.277906	0.001239	0.5	0	0.000213	1.779357	Yes
n	1.277906	0.000619	1.0	0	0.000213	2.278738	Yes
$n/0.5$	1.277906	0.000310	2.0	0	0.000213	3.278428	No
$n/0.1$	1.277906	0.000062	10.0	0	0.000213	11.278180	No



(a) Generated samples



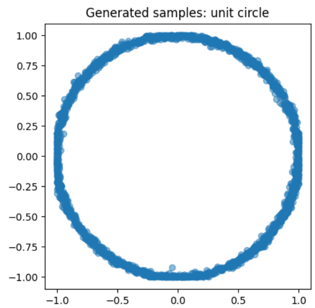
(b) Real samples

Figure 3: Comparison between generated and real samples for the circle distribution

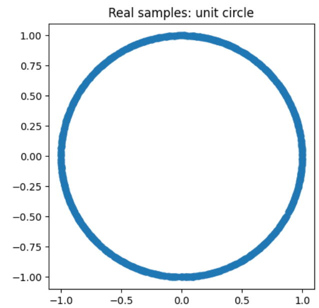
3.2.1.1 Circle Result 1

Table 5: Bound terms for the unit-circle dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.275073	0.006193	0.1	0	0.000213	1.381480	Yes
$n/5$	1.275073	0.003097	0.2	0	0.000213	1.478383	Yes
$n/2$	1.275073	0.001239	0.5	0	0.000213	1.776525	Yes
n	1.275073	0.000619	1.0	0	0.000213	2.275906	Yes
$n/0.5$	1.275073	0.000310	2.0	0	0.000213	3.275596	No
$n/0.1$	1.275073	0.000062	10.0	0	0.000213	11.275348	No



(a) Generated samples



(b) Real samples

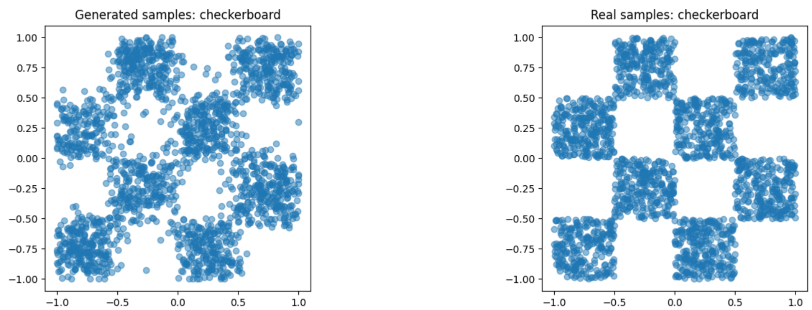
Figure 4: Comparison between generated and real samples for the circle distribution

3.2.1.2 Circle Result 2

3.2.2 Checkerboard

Table 6: Bound terms for the checkerboard dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.047146	0.006125	0.1	0	0.000213	1.153484	Yes
$n/5$	1.047146	0.003063	0.2	0	0.000213	1.250421	Yes
$n/2$	1.047146	0.001225	0.5	0	0.000213	1.548583	Yes
n	1.047146	0.000613	1.0	0	0.000213	2.047971	Yes
$n/0.5$	1.047146	0.000306	2.0	0	0.000213	3.047665	No
$n/0.1$	1.047146	0.000061	10.0	0	0.000213	11.047420	No



(a) Generated samples

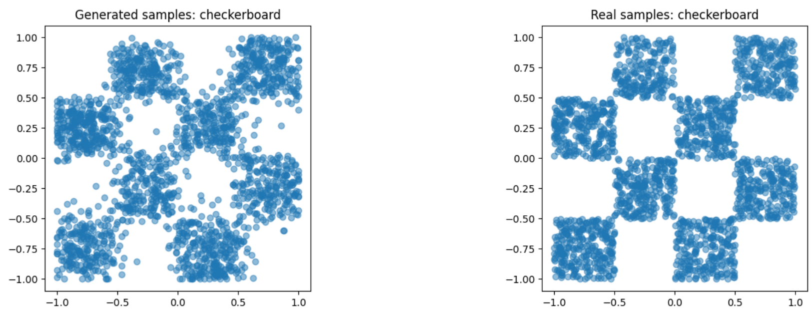
(b) Real samples

Figure 5: Comparison between generated and real samples for the checkerboard

3.2.2.1 Checkerboard Result 1

Table 7: Bound terms for the checkerboard dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.036882	0.006126	0.1	0	0.000213	1.143221	Yes
$n/5$	1.036882	0.003063	0.2	0	0.000213	1.240158	Yes
$n/2$	1.036882	0.001225	0.5	0	0.000213	1.538320	Yes
n	1.036882	0.000613	1.0	0	0.000213	2.037707	Yes
$n/0.5$	1.036882	0.000306	2.0	0	0.000213	3.037401	No
$n/0.1$	1.036882	0.000061	10.0	0	0.000213	11.037156	No



(a) Generated samples (b) Real samples

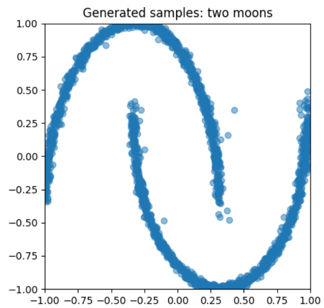
Figure 6: Comparison between generated and real samples for the checkerboard

3.2.2.2 Checkerboard Result 2

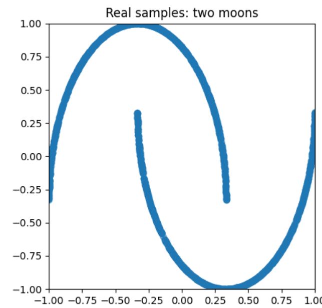
3.2.3 TwoMoons

Table 8: Bound terms for the two-moons dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.093001	0.006146	0.1	0	0.000213	1.199359	Yes
$n/5$	1.093001	0.003073	0.2	0	0.000213	1.296286	Yes
$n/2$	1.093001	0.001229	0.5	0	0.000213	1.594442	Yes
n	1.093001	0.000615	1.0	0	0.000213	2.093828	Yes
$n/0.5$	1.093001	0.000307	2.0	0	0.000213	3.093520	No
$n/0.1$	1.093001	0.000061	10.0	0	0.000213	11.093275	No



(a) Generated samples



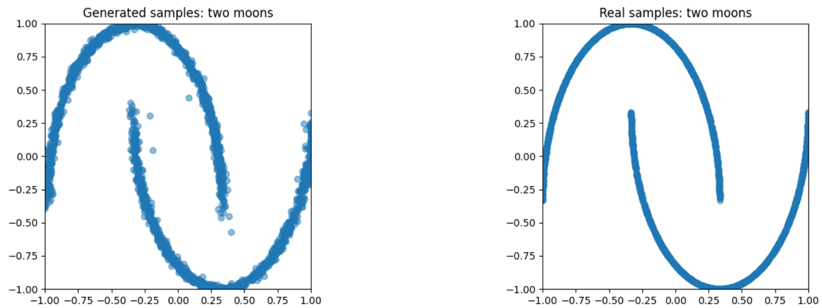
(b) Real samples

Figure 7: Comparison between generated and real samples for the two-moons distribution

3.2.3.1 TwoMoons Result 1

Table 9: Bound terms for the two-moons dataset for different values of λ.

λ	Reconstruction	Prior matching and confidence	Diameter term	Terminal mismatch	Accumulated noise	Total bound	Below $\sqrt{8}$
$n/10$	1.100248	0.006145	0.1	0	0.000213	1.206606	Yes
$n/5$	1.100248	0.003072	0.2	0	0.000213	1.303533	Yes
$n/2$	1.100248	0.001229	0.5	0	0.000213	1.601690	Yes
n	1.100248	0.000614	1.0	0	0.000213	2.101075	Yes
$n/0.5$	1.100248	0.000307	2.0	0	0.000213	3.100768	No
$n/0.1$	1.100248	0.000061	10.0	0	0.000213	11.100522	No



(a) Generated samples

(b) Real samples

Figure 8: Comparison between generated and real samples for the two-moons distribution

3.2.3.2 TwoMoons Result 2

3.3 Discussion

3.3.1 Evaluation of the replication

The replication results show close agreement with the numerical values reported by Mbacke and Rivasplata (2024). In the first run, the relative differences across the six values of λ ranged from 0.007% to 3.344%, while in the second they ranged from 0.019% to 3.313%. The largest differences occurred at $\lambda = n/10$, whereas the results for the two largest values of λ were particularly close to the published values. More importantly, both runs reproduced the same overall behaviour across λ: the bound increased as λ increased, remained below the trivial upper bound of $\sqrt{8}$ for $\lambda \in \{n/10, n/5, n/2, n\}$, and exceeded it for $\lambda = n/0.5$ and $\lambda = n/0.1$. The replication therefore reproduced both the approximate numerical values and the informative-to-vacuous transition observed in the original experiment.

The two independent runs were also highly consistent with one another. For each value of λ, the difference between the two replicated total bounds was approximately 3.5×10^{-4}. Given that the training and certificate evaluation contain stochastic components, this small between-run variation indicates that the observed numerical pattern was repeatable across the two experiments. This conclusion should nevertheless be interpreted cautiously: two runs are sufficient to show that the reported result was not unique to a single run, but they are not sufficient to characterise the full variability of the procedure.

The generated samples provide complementary qualitative evidence. In both replications, the model-generated points broadly reproduce the spatial coverage of the uniform-square target distribution. This visual agreement is not used as evidence for the validity of the certificate itself, since visual inspection does not quantify distributional discrepancy, but it confirms that the replicated model learned the expected overall target structure.

Overall, the replication reproduced the numerical behaviour reported in the original experiment with only small differences and showed little variation between the two independent runs. These results provide the reference point for evaluating how the same certificate behaves under the alternative target distributions considered in the extension experiments.

3.3.2 Evaluation Across Target Distributions

The extension experiments first examine whether the certificate remains informative when the original uniform-square target is replaced by distributions with different support structures. All experiments use the common instance space $\mathcal{X} = [-1, 1]^2$, giving the same diameter $\Delta = \sqrt{8} \approx 2.828$. For circle, checkerboard and two moons, both runs produced certificates below this baseline for $\lambda \in \{n/10, n/5, n/2, n\}$, while the two larger settings, $\lambda = n/0.5$ and $\lambda = n/0.1$, exceeded it. The method therefore continued to produce non-vacuous numerical certificates across all three additional distributions for the first four parameter settings.

The cross-distribution differences in the total certificate are concentrated almost entirely in the reconstruction term. Averaged across the two runs, the reconstruction values are approximately 1.042 for checkerboard, 1.055 for square, 1.097 for two moons and 1.276 for circle. In comparison, the diameter term is identical across datasets at any fixed λ, since it depends only on λ, n and the common Δ; the prior-matching and confidence contribution varies only slightly; and the accumulated-noise term remains approximately 2.13×10^{-4}. The terminal-mismatch term is displayed as zero at every reported precision. This reflects the diffusion schedule rather than the evaluation procedure, since the term carries the factor $\prod_{t=1}^{T} K'_t$ with $T = 1000$ and $K'_t < 1$ at every step, placing the product far below double-precision range. The differences in total bound across datasets therefore reflect variation in a single component, whose behaviour is examined below.

This produces a consistent ordering of checkerboard < square < two moons < circle at matched values of λ, and the ordering does not correspond to a simple notion of geometric complexity. Checkerboard introduces fragmentation and multimodality yet produces the smallest reconstruction term, within 0.013 of the connected, unimodal square, whereas circle has the visually simplest structure of the four yet exceeds checkerboard by 0.23.

A scalar property of the target predicts the ordering directly. Because the linear schedule gives $\bar{\alpha}_T \approx 4 \times 10^{-5}$ at $T = 1000$, the terminal distribution $q(x_T \mid x_0)$ retains almost no information about x_0: its mean is displaced by less than 0.7% of x_0 in a distribution of unit scale. A trajectory started from such an x_T is therefore close to an independent draw from π_θ, and the reconstruction term measures the expected distance between an independent target sample and an independent generated sample rather than the fidelity of any particular reconstruction. For a well-fitted model this approaches the mean pairwise distance of μ itself, $E_{x,y\sim\mu}\|x - y\|$, a constant determined by the target alone and computable before any model is trained. Table 10 compares that constant, estimated by pairing independent draws, with the measured reconstruction terms.

Table 10: Mean pairwise distance of each target distribution compared with the measured reconstruction term.

Target	$E_{x,y\sim\mu}\|x - y\|$	Reconstruction	Difference	Relative (%)
Checkerboard	1.0394	1.042	+0.003	0.25
Square	1.0426	1.055	+0.012	1.19
Two moons	1.1004	1.097	−0.003	0.31
Circle	1.2733	1.276	+0.003	0.21

The two quantities agree to within 1.2% across all four targets, and the ordering is reproduced exactly, including the narrow separation between checkerboard and square. For circle the estimate also matches the analytical value $4/\pi \approx 1.2732$. Three of the four residuals are approximately 0.003 with mixed sign, consistent with Monte Carlo variation rather than systematic model misfit; the exception is the square, whose residual of 0.012 is four times larger and which is also the only dataset trained under the unmodified DataLoader.

Two consequences follow. First, the circle yields the loosest certificate not because it is harder to fit but because points on a circumference of radius one are on average further apart than points drawn from the square inscribing it; the apparent association with support dimensionality is a consequence of this rather than an explanation in its own right, since removing a dimension of the support also changes the mean pairwise distance. Second, and more consequentially, the dominant component of the certificate is in this regime determined by a property of the target distribution to a precision at which the contribution of the trained model is not resolvable. The certificate remains below $\sqrt{8}$ across these datasets substantially because their mean pairwise distances, ranging from 1.04 to 1.27, are themselves well below $\sqrt{8}$. This qualifies the sense in which the method extends to new targets: it does so reliably, but the numerical value obtained is largely a statement about μ rather than about π_θ.

Circle and two moons also extend the numerical evaluation to a setting explicitly covered by the theory but not demonstrated in the original experiment. Mbacke and Rivasplata (2024) state that the bound applies to bounded target distributions even when they do not admit a density with respect to Lebesgue measure, while their reported numerical experiment considers only the full-dimensional uniform square. The noise-free circle and two-moons distributions are supported on one-dimensional subsets of R^2, and both still yield non-vacuous certificates. These experiments therefore provide a numerical illustration of the certificate in a class of distributions included in the theoretical result but absent from the original empirical evaluation.

Three limitations qualify these conclusions. All four targets remain two-dimensional and share a single bounded instance space; holding Δ fixed is what makes the comparison interpretable, but it also means the certificate was never tested against a change in dimensionality or diameter. The square model was trained under a different effective batch size from the three extension models, so its residual in Table 10 is not directly comparable with theirs. Finally, each reconstruction term rests on a Monte Carlo estimate using one terminal draw and one reverse trajectory per certification sample, and its variance was not characterised; the agreement reported in Table 10 is therefore consistent with, but does not by itself establish, an exact identity between the two quantities.

3.3.3 Analysis of Bound Components

The decomposition of the certificate reveals a consistent pattern across all target distributions. While the reconstruction term determines most of the cross-distribution variation discussed in Section 3.3.2, the changes in the total bound across different values of λ are driven primarily by the diameter term. The remaining components make comparatively small contributions under the present experimental configuration.

The prior-matching and confidence term decreases as λ increases, consistent with its $1/\lambda$ dependence in Theorem 3.1. Across the extension experiments, this contribution decreases from approximately 0.006 at $\lambda = n/10$ to around 6×10^{-5} at $\lambda = n/0.1$. Although this represents a substantial proportional reduction, the term is already small relative to the reconstruction component and therefore has little influence on the overall certificate.

By contrast, the diameter penalty increases linearly with λ. Since the experiments use $n = 5000$ and $\Delta = \sqrt{8}$,

$$\frac{\lambda\Delta^2}{8n} = \frac{\lambda}{n},$$

giving contributions of 0.1, 0.2, 0.5, 1, 2, and 10 for the six values of λ. This increase is much larger than the simultaneous reduction in the prior-matching term. Consequently, the diameter contribution becomes progressively more important and is the principal reason that the certificate exceeds the trivial $\sqrt{8}$ baseline at $\lambda = n/0.5$ and $\lambda = n/0.1$. The loss of non-vacuity at larger λ therefore reflects the structure of the bound rather than a deterioration in model reconstruction performance.

The two remaining propagation-related components are negligible in comparison. The terminal-mismatch term evaluates numerically to zero in the reported experiments, while the accumulated-noise contribution remains approximately 2.13×10^{-4}. As a result, the certificate in the present setting is numerically dominated by the reconstruction and diameter components.

This decomposition clarifies two distinct sources of variation in the results. Differences between target distributions are expressed mainly through the reconstruction term, whereas differences across λ are dominated by the diameter penalty. Although λ theoretically controls a trade-off between prior matching and the diameter term, that trade-off is highly asymmetric in these experiments because the prior-matching contribution is already very small. Among the six values evaluated, $\lambda = n/10$ therefore produces the smallest certificate for every distribution; this should be interpreted only within the tested parameter range rather than as evidence that $n/10$ is globally optimal.

Consequently, over the range examined here, increasing λ consistently loosens the certificate and ultimately causes it to become vacuous, directly answering RQ2 regarding the numerical effect of the trade-off parameter.

3.3.4 Numerical and Qualitative Evaluation

The evaluation methods were selected to align directly with the project goals and research questions. The Wasserstein certificate is the primary quantitative measure because RQ3 asks whether the method remains informative when extended to alternative target distributions; comparing the calculated bound with the common baseline of $\sqrt{8}$ provides a direct assessment of non-vacuity. The bound decomposition additionally addresses RQ2 by showing how λ affects the numerical behaviour of the certificate. Generated-sample plots provide complementary qualitative evidence that the trained models recover the intended target structures.

The plots also show that certificate tightness does not directly correspond to apparent generation quality. Circle visually reproduces the target structure closely, with generated samples concentrated around the circumference, yet it has the largest certificate among the extensions. Checkerboard shows more visible leakage into nominally empty regions and softer boundaries while producing the smallest certificate. Two moons similarly recovers both curved components, with some dispersion around the noise-free support.

This contrast is consistent with Section 3.3.2, where the reconstruction term was found to be closely associated with the target distribution's mean pairwise distance. Visual inspection and the certificate therefore capture different aspects of model behaviour: the plots assess qualitative recovery of the target structure, while the certificate provides the formal numerical evaluation required to answer RQ2 and RQ3. Neither is treated as a substitute for the other.

3.3.5 Critical Reflection and Limitations

A principal strength of the extension is its controlled experimental design. The target distributions were varied while the ambient dimensionality, bounded instance space, diffusion architecture, training configuration and certificate-evaluation procedure were kept largely fixed. This makes the comparison between datasets easier to interpret, since changes in the bound can be related more directly to changes in the target distribution rather than to simultaneous changes in model scale or theoretical assumptions. The dataset selection was also theoretically motivated rather than arbitrary: circle and two moons introduce noise-free lower-dimensional supports without a two-dimensional Lebesgue density, allowing the numerical evaluation to cover a class of distributions explicitly permitted by the theory but not examined in the original experiment.

The same controlled design nevertheless limits the scope of the conclusions. All extension experiments remain two-dimensional and synthetic. Although circle, checkerboard and two moons introduce more varied geometry than the uniform square, they remain substantially simpler than high-dimensional generative-modelling tasks such as MNIST or CIFAR. The present results therefore demonstrate extensibility across structurally different low-dimensional distributions, but do not establish that the certificate will remain numerically informative as dimensionality and model complexity increase. Similarly, retaining the common instance space $\mathcal{X} = [-1,1]^2$ and fixed diameter $\Delta = \sqrt{8}$ strengthens the internal comparison, but means that the sensitivity of the certificate to changes in instance-space diameter was not tested. A further asymmetry is that the square model retains the uncorrected DataLoader behaviour of the released implementation, giving it an effective batch size of one, whereas the three extension models were trained with the intended batch size. Its residual is therefore not directly comparable with theirs.

A further strength is that the evaluation goes beyond reporting the final certificate value. Decomposing the bound identifies which terms are responsible for variation across datasets and across λ, while the generated-sample comparisons provide complementary evidence about whether the intended target structures were learned. This analysis also revealed an important limitation in how non-vacuity should be interpreted. A certificate below the trivial $\sqrt{8}$ baseline is informative relative to that baseline, but it does not necessarily imply that the bound is tight or that it closely reflects generative model quality. In the present experiments, the dominant reconstruction component was closely matched by the mean pairwise distance of the target distribution itself. This indicates that a substantial part of the numerical certificate can be explained by properties of μ, rather than by differences in the quality of the learned distribution π_θ. The qualitative sample comparisons reinforce this distinction, since the ordering of certificate values does not directly follow the apparent quality of the generated samples.

Finally, the numerical robustness of the procedure was only partially assessed. Two independent runs were sufficient to show that the main patterns were repeatable, but they do not characterise the full variability introduced by stochastic training and Monte Carlo certificate estimation. In particular, each reconstruction estimate uses one terminal draw and one reverse trajectory per certification sample, and the variance of this estimator was not quantified. A more extensive evaluation could therefore use additional independent training runs, repeated reverse trajectories and explicit uncertainty estimates. Future work could also test whether the certificate responds meaningfully to model quality by comparing checkpoints or deliberately degraded models, and could extend the analysis to higher-dimensional datasets to assess whether the observed non-vacuity persists beyond the controlled two-dimensional setting.

Taken together, these strengths and limitations define the scope of the present findings. The experiments provide a controlled and theoretically motivated evaluation of reproducibility, non-vacuity and bound composition across several distinct target distributions, while leaving open the broader questions of certificate tightness, model sensitivity and scalability to realistic high-dimensional generative tasks.

4 Conclusion

4.1 Conclusion

The project achieved its main objectives through replication, analysis of the trade-off parameter λ, and controlled extension to alternative target distributions. These outcomes provide direct answers to the three research questions.

For RQ1, the published Wasserstein certificate values were reproduced with only small numerical differences across two independent runs, including the same transition from informative to vacuous bounds. This supports the reproducibility of the original numerical experiment.

For RQ2, λ was found to control a strongly asymmetric trade-off. Increasing λ reduced the prior-matching and confidence term, but this reduction was outweighed by the increasing diameter penalty. Consequently, larger tested values of λ produced progressively looser certificates and eventually caused the bound to become vacuous.

For RQ3, the certification procedure was successfully extended to circle, checkerboard and two-moons distributions while retaining the controlled bounded setting. All three produced non-vacuous certificates for the first four tested values of λ, including the two lower-dimensional distributions without a two-dimensional Lebesgue density. However, the extension also showed that non-vacuity should not be interpreted directly as certificate tightness or model quality, since the dominant reconstruction component was strongly influenced by properties of the target distribution itself.

Overall, the project fulfilled the proposed replication and extension objectives while also identifying important questions concerning the interpretation and broader applicability of the certificate.

4.2 Future Work

Three directions follow from the present findings. First, the extensions remain limited to two-dimensional synthetic distributions. Future work could move to higher-dimensional data, progressing from controlled synthetic settings to datasets such as MNIST, with a more suitable diffusion architecture.

Second, the relationship between certificate tightness and model quality should be tested directly. Since the reconstruction term was strongly influenced by properties of the target distribution, future experiments could hold the target fixed while varying model quality through different checkpoints, training durations or model capacities.

Third, the numerical robustness of the certificate could be examined more thoroughly. Additional independent runs would allow variability due to stochastic training and data sampling to be characterised more reliably, which is important when interpreting empirical differences between learning procedures (Bouthillier et al., 2021). Repeated terminal draws and reverse trajectories could similarly reduce Monte Carlo uncertainty, since the standard error of conventional Monte Carlo estimation decreases at the $O(N^{-1/2})$ rate as the number of independent samples increases (Caflisch, 1998). A denser evaluation of λ could also provide a more complete characterisation of the trade-off between the decreasing prior-matching term and increasing diameter penalty.

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